

Seaglider refurbishment training

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This document provides an overview of Seaglider refurbishment (inspection, battery swap, CT sail removal, and basic ballasting). It assumes basic knowledge of Seaglider principals and operation.

If you have any reason to suspect water intrusion into the glider (e.g., if there is a visible hull penetration, history of very rough handling, or unusually high internal pressure and/or relative humidity), proceed with extreme caution as the lithium batteries may be compromised:

- Wear PPE (eye protection, gloves)
- Store and open the glider outdoors
- Keep your face and body away from the glider as you are opening it
- Have Lith-X or a lithium fire extinguisher available

I. Materials and general comments

Common Seaglider o-ring sizes (all are buna rubber o-rings):

Size	Use
012	CT sail bore seal (also uses an 012 plastic back-up ring) Antenna bulkhead connector
013	DGO connector
016	CT sail face seal
263	Electronics hull (both ends)
270	Aft endcap to battery hull, bulkhead to battery hull
271	Battery hull to joint ring
5-712 or N70- M1.3x7.95	IE-55 bulkhead connectors

Common chemicals and spare parts:

Name	Use
Loctite 243 (blue)	Threads of all fasteners that are not exposed to seawater
Tef-gel	- Threads of fasteners that are exposed to seawater (hull sections, fairings) - Face of bulkhead connectors - Bladder nut threads
Acetone	Great for cleaning hydraulic oil. Do not use on rubber parts, wings/rudder
Isopropyl alcohol	Less effective and less toxic than acetone
Chemtrinic Max-Kleen	Cleaner for IE-55 connectors
3M silicone spray lubricant	Lubricant for connectors
Zinc anode	https://www.fisheriessupply.com/martyr-marine-engine-pencil-anodes-only-zinc/cmez1

General guidance on fasteners

- Use blue loctite on threads of all fasteners, except fasteners that are exposed to seawater
- For fasteners that are exposed to seawater, apply Tef-gel to threads
- If any fasteners come loose *at all* during the refurb process, remove them completely and re-apply Loctite
- Do not reuse fasteners that are corroded, bent or damaged. After long deployments, consider replacing all fasteners.

II. Inspection

Visually inspect the glider for any obvious damage to fairings, rudder/wings, antenna, sensors, cables, etc.

Before opening the glider, it is useful to gather baseline data about the hardware, electronics, and sensors. This helps with identifying problems and also provides a basis for comparison once the glider is reassembled. This can be done by performing a selftest (or a nosed selftest if you are indoors - this is identical to a typical auto selftest except that it doesn't include the GPS reading or phone call) or running through the hardware menu as described here.

With the glider on the bench or in a jig, connect to the glider with the comms cable, open a terminal window, and power on. Run through the hardware (hw) menu and test all systems:

(1) hw/pitch

- Move the pitch mass all the way forward, then all the back, then forward again.
- Listen for any unusual sounds from the motor
- Look at the output of the motor moves: current draw (look for very high or low values), pitch AD position (look for jumps or hangups), time to complete the move (look for very short or long motor moves). Note the motor rates and currents (baseline data to compare to later in the refurb process)
- Leave the pitch motor at ~2000 AD counts so that the ribbon cables aren't damaged during the roll moves

(2) hw/roll

- Move the roll mass all the way to one side, then all the way to the other side
- Listen for any unusual sounds from the motor
- Look at the output of the motor moves (current rates, AD position, time) and note the motor rates and currents
- Move the roll back to center

(3) hw/vbd

- Bleed the VBD all the way down, then pump to max
- Listen for any unusual sounds from the motor
- Look at the output of the motor moves (current rates, AD position, time) and note the motor rates and currents
- Bleed most of the oil so that access to the endcap and connectors will be easier

(4) hw/pressure

- run a sealevel test. Large changes in y-intercept (> a few psi) compared to previous tests, or large standard deviations, suggest a problem with the pressure sensor

(5) hw/compass/dispraw

- verify that compass data are reasonable (since the glider isn't moving, data will not be accurate)

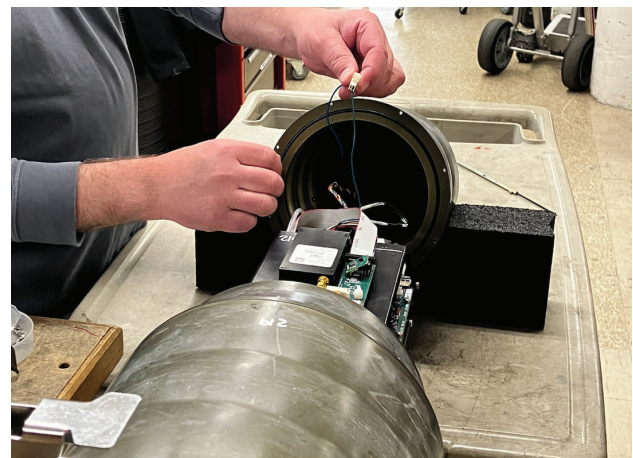
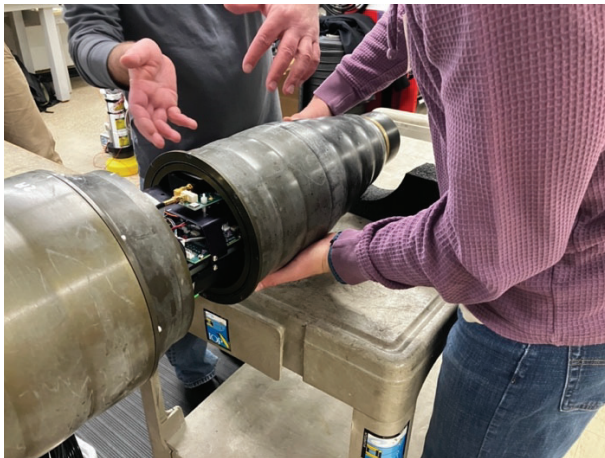
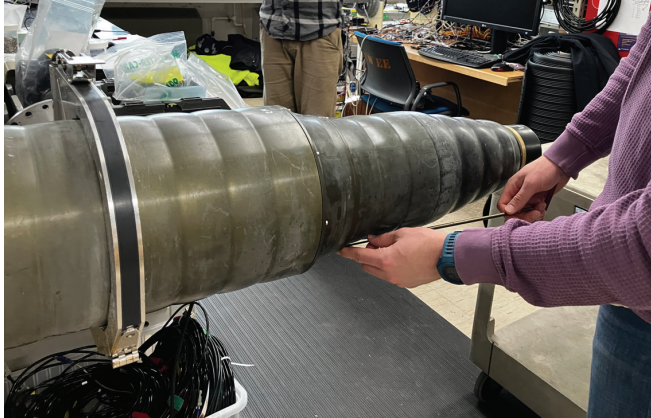
(6) hw/gps/display

- verify that GPS data are being read (GPS data will not be accurate without antenna attached and clear sky view)
- (7) hw/modem/id**
 - verify that the output is correct (IMEI, SIM)
- (8) hw/intpress/selftest**
 - check that the glider is still under vacuum internal pressure (~8.5 psi). If internal pressure is closer to atmospheric pressure (~14.5 psi), the glider has lost vacuum and may have water intrusion
 - Similarly, if RH is very high (> 70%), there may be water intrusion
 - If internal pressure or RH values make you suspect water intrusion, assume the battery may be exposed to water and take all safety/PPE precautions (see above)
- (9) hw/altim/ping**
 - verify that the altimeter is pinging (listen for chirp)
- (10) hw/sensors**
 - run through all attached sensors and verify they are connected
 - if running SciCon, sensors will not appear here
- (11) hw/loggers**
 - if running SciCon: run a selftest from the scicon menu and verify that the output is as expected (instrument, attach, and scheme files are present and correct)
- (12) hw/batt/read**
 - check the fuel gauge reading. If reusing the batteries, note this number (e.g., note it on a piece of tape and stick to the battery packs)
 - reset the battery numbers
- (13) shutdown** (can be done from the main menu on RevE gliders)

III. Glider disassembly

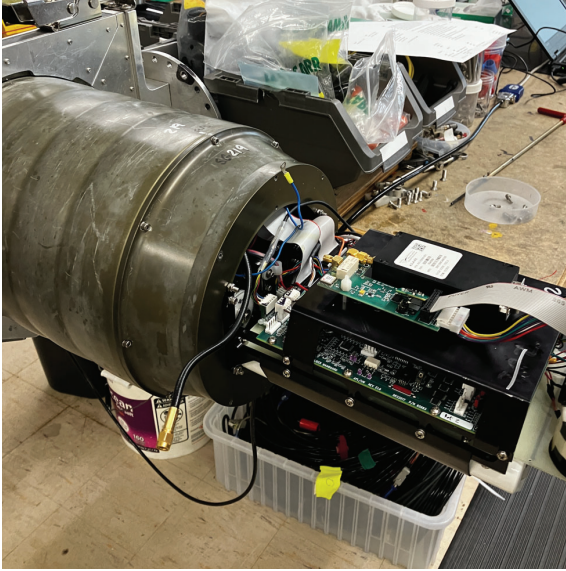
With the glider powered off, disassemble it section by section. If the CT sail or VBD are not being removed, it is only necessary to disassemble as far as the mass shifter/main battery (steps 1-9). Reassembly will be easier if you take lots of photos during disassembly.

- (1) remove forward fairing and affix the glider in jig
- (2) disconnect and remove sensors and hatch cover
 - put dust cap on the antenna cable and bulkhead to avoid losing the o-ring
- (3) remove aft fairing
- (4) with the glider horizontal in the jig, use the PRV tool to release the vacuum
 - listen for the hissing sound
 - screw the PRV back in, ensuring that the o-ring is still in place
- (5) remove the forward hull section:
 - remove fasteners and gently slide the nose straight out and onto a cart/table
 - disconnect grounding strap, compass, and transducer wires, then set nose aside

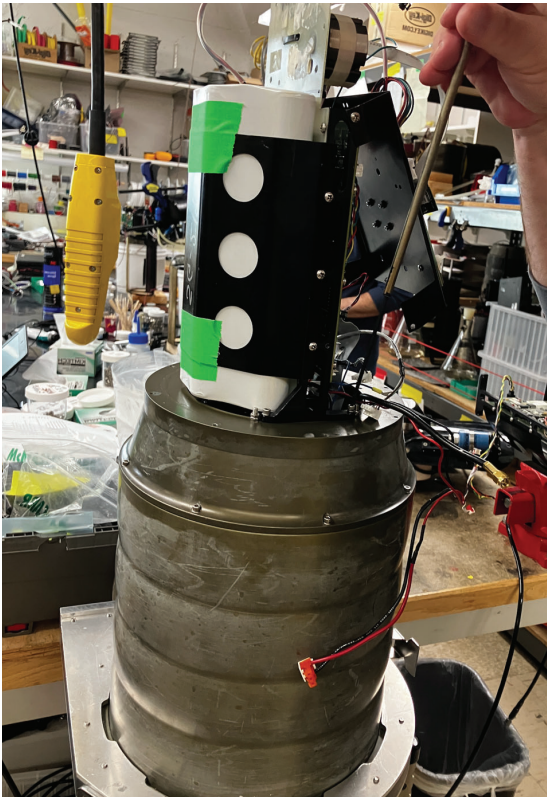


(6) disconnect electronics:

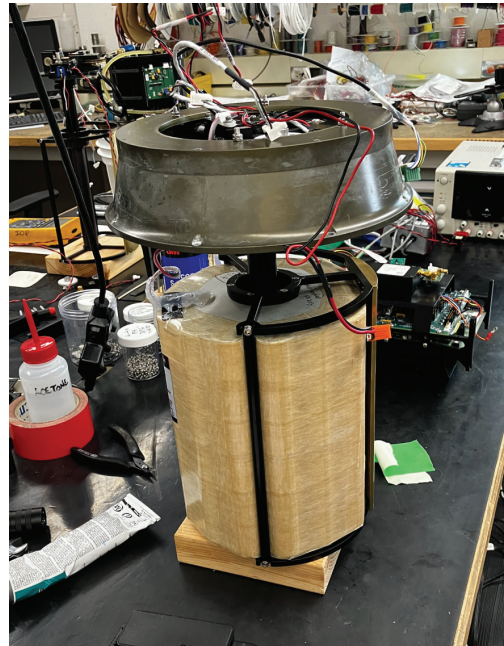
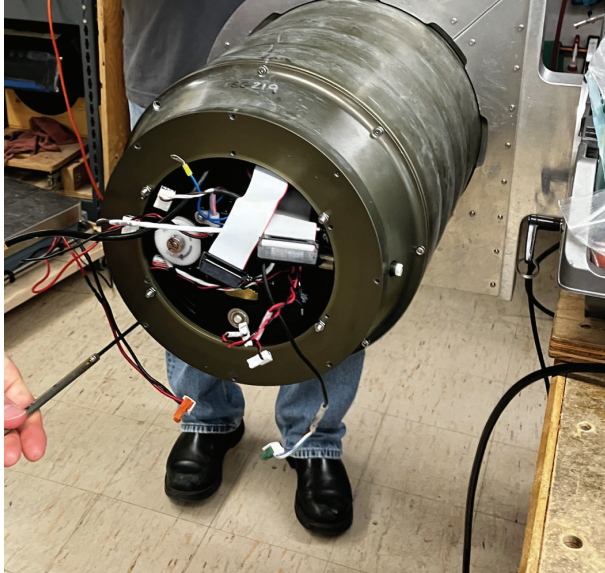
- disconnect the coax cable and grounding strap from the motherboard
- tip back the phone bridge and disconnect the ribbon cables, batteries, and potentiometers
- motherboard can typically be left in place for a standard refurb; if necessary, the electronics can be dropped by removing the 4 fasteners connecting the board to the rails (do this with the glider vertical)



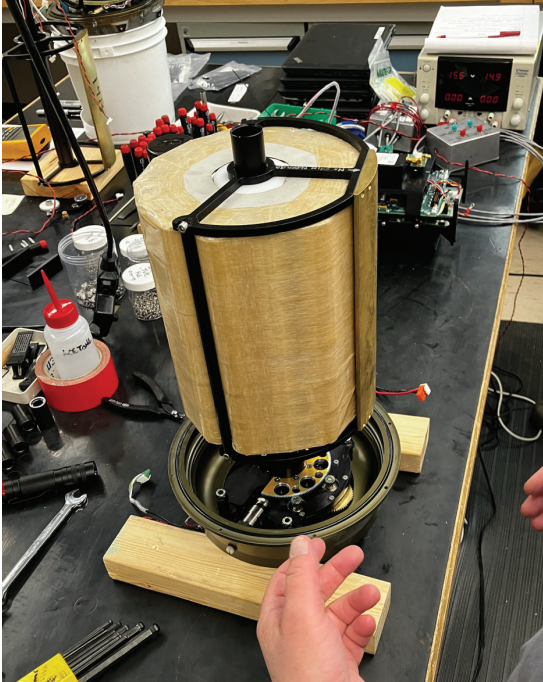
- (7) tilt the glider vertical and unfasten/remove the entire electronics/forward battery assembly
- untape and remove the forward battery pack from the assembly. It should just slide out, but loosen fasteners if needed



- (8) with the glider horizontal, unfasten and remove the bulkhead section
- slide the mass shifter out, being careful that the cables don't get snagged
 - rest the mass shifter assembly, battery pack down, supported on a block with a hole in the center or across two blocks

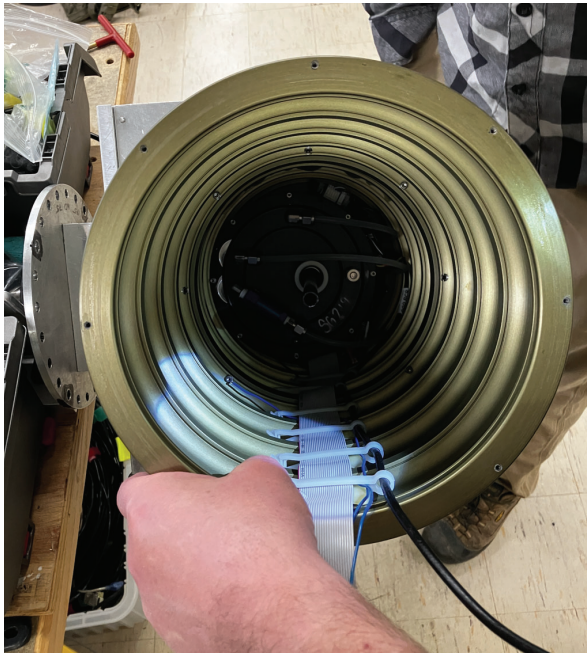


- (9) Disconnect the battery cable. With the battery up, remove the plate on top of the battery. Loosen the vertical rails holding the battery in place and remove the battery from the assembly



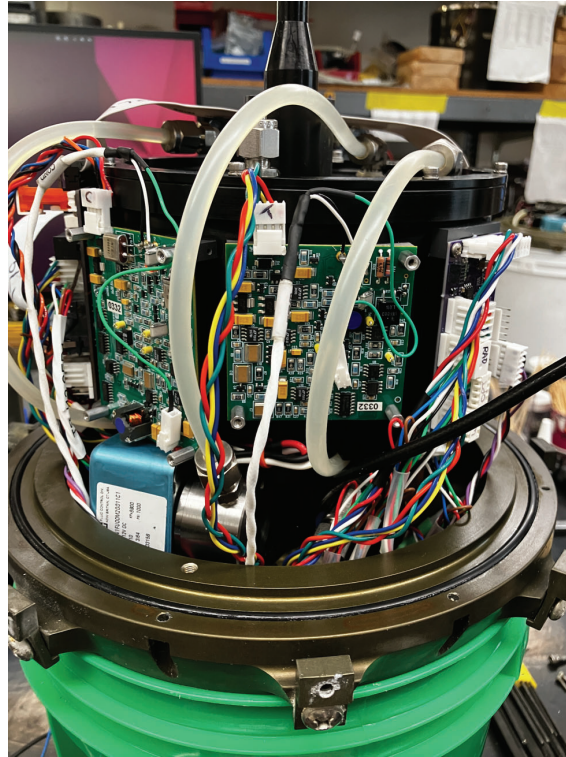
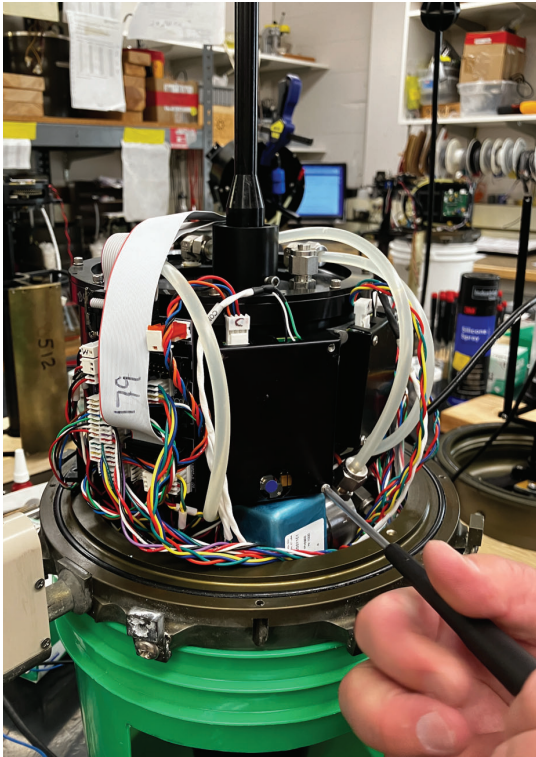
(10) Aft endcap:

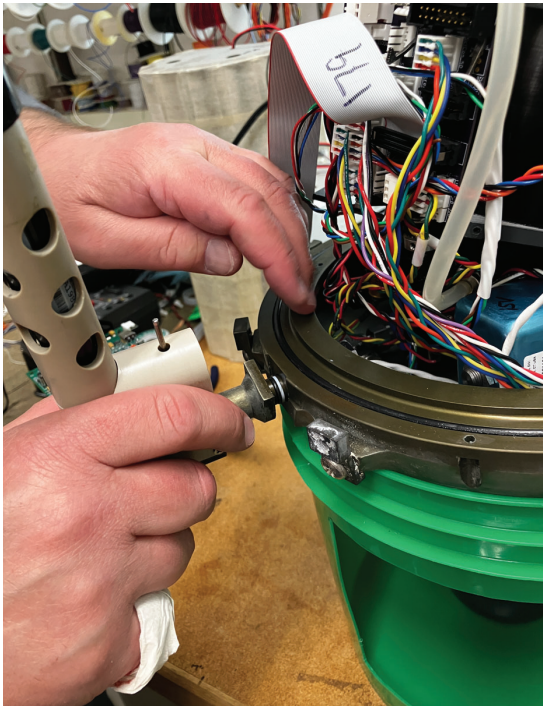
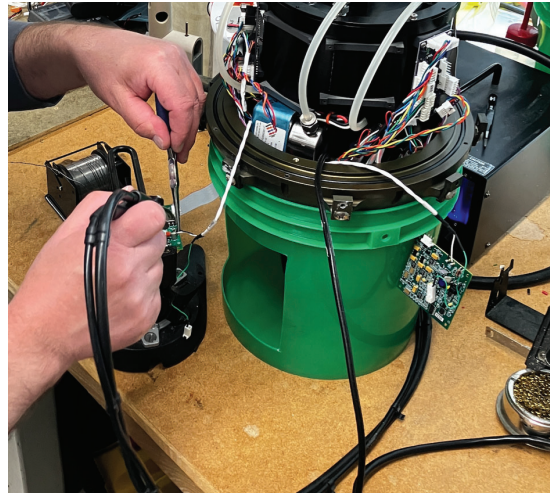
- Mark the orientation of the endcap where it meets the adjoining hull section so it's easier to align during reassembly
- Tilt the glider so it is facing upward at a 45° angle
- Before removing, it can be helpful to mark the coax cable where emerges from the hull so that it more easily positioned during reassembly
- Unclip all cables
- Flip the glider vertical, aft side up, and inspect the endcap
- Carefully inspect and clean all IE-55 bulkheads and dummies. We clean them with Chemtronics Max-Kleen degreaser and a toothbrush. Inspect carefully for damage to rubber or pins.
- If leaks or wear/damage to IE-55s is suspected, the IE-55 bulkheads can be removed. Inspect the IE-55 ports on the endcap, looking for corrosion or damage. Clean the o-ring surface and the underside of the connector with isopropyl and reinstall: grease and replace the o-ring, use blue Loctite on the IE-55 threads and tef-gel on the connector face. Do not overtighten.
- Mark the position of the CT sail on the main hull section so that the aft endcap can be easily aligned during reinstallation
- Remove the aft endcap and carefully guide cables through the hull. Set aft endcap in a bucket.
- Remove the DGO bulkhead connector on the aft endcap: inspect carefully for any signs of corrosion. Minor corrosion should be monitored; in the case of major corrosion / pitting, the endcap may need to be replaced.
- Carefully reinstall the DGO with loctite on the threads, o-ring grease on the o-ring, and generous tef-gel on the face. Do not overtighten!
- Remove the zinc anode (carefully – they can break off easily) and replace with a new one



(11) Remove CT sail:

- Remove conductivity and temperature PCB covers. Take note of how the leads are installed – for conductivity, white wire typically goes to the positive terminal and black wire to the negative
- Unfasten the PCBs so they can be (de)soldered more easily
- Desolder the wires from both boards
- Unfasten the CT sail nut from the inside of the endcap and carefully slide out the CT sail and wires
- Ship the CT sail and boards, and a CT calibrating housing, to Seabird for calibration. Ship with the boards in anti-static bags, wrapped in bubble wrap.



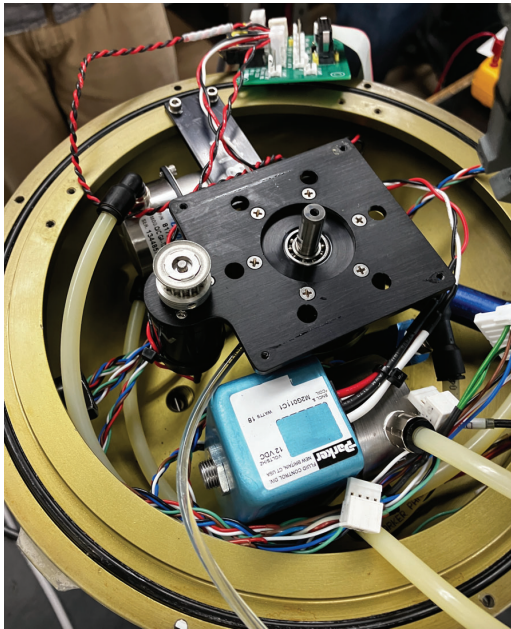


To reinstall the calibrated CT sail:

- Resolder the two PCBs, taking care to attach the wires to the correct positive/negative terminals. Do not clip the wires – their length matters for the calibration
- Re-attach the CT sail to the endcap and affix with the CT sail nut. Tighten down until you feel metal on metal
- The hull is keyed so that the CT sail will be correctly aligned
- Attach the grounding wire (use loctite)

(12) VBD endcap

- Use a flashlight to inspect the endcap carefully for any signs of oil. Look closely near fittings, around the bottom of the pump, at the bottom of the endcap for oil that may have pooled
- Carefully clean the endcap with acetone so that any oil observed in the future can be identified
- Shake the endcap: an audible sloshing sound indicates air in the system. In this case, back off the bleed screw slowly and carefully (do NOT remove all the way. It is only 3/8 long) and jiggle the screw until you see air starting to bubble out. Tilt the endcap slightly so that the bleed screw is the highest point and air can more easily escape. Retighten the bleed screw and shake the endcap; repeat until you no longer hear sloshing.



IV. Reassembly

(1) O-rings

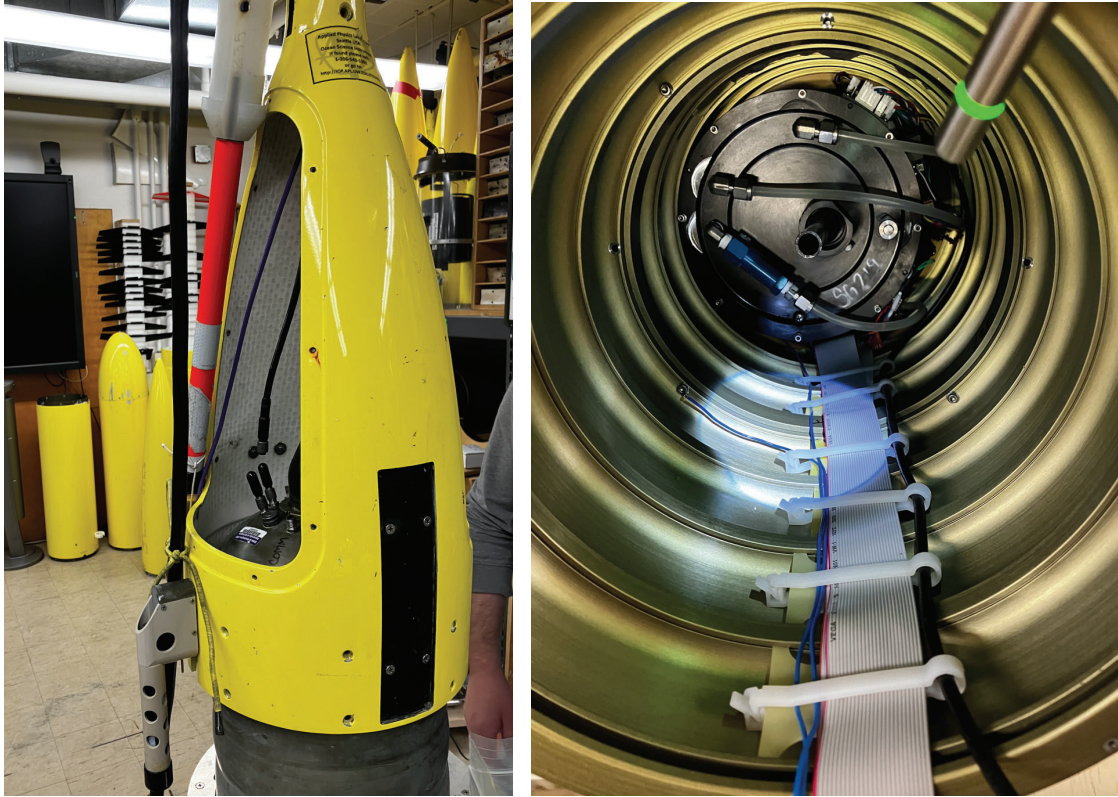
- Carefully clean all o-ring grooves and faces and replace all o-rings between the hull sections
- If the inside of the hull is oily, clean with acetone

(2) Reattach the aft endcap

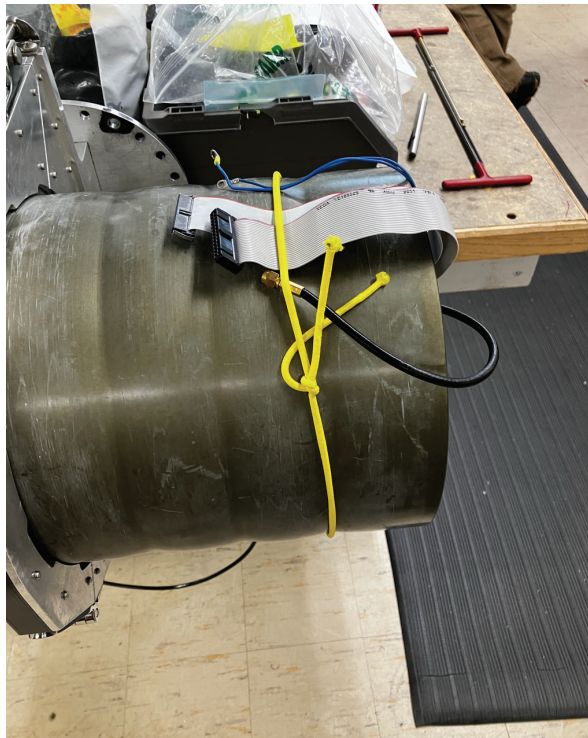
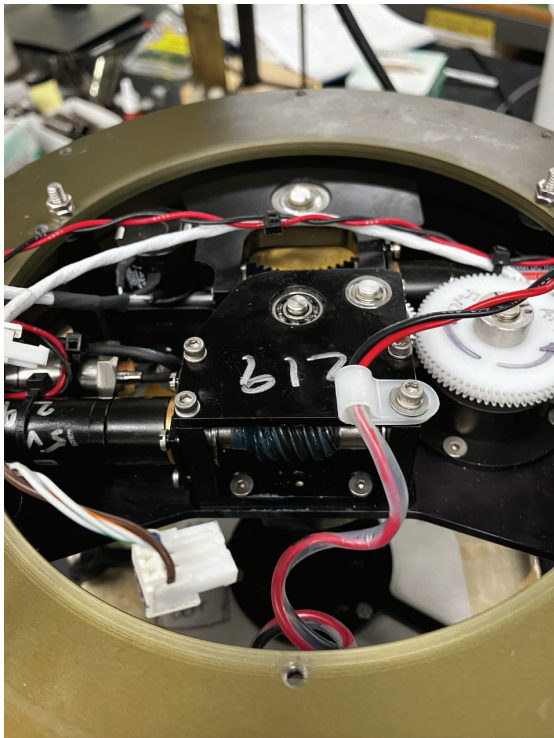
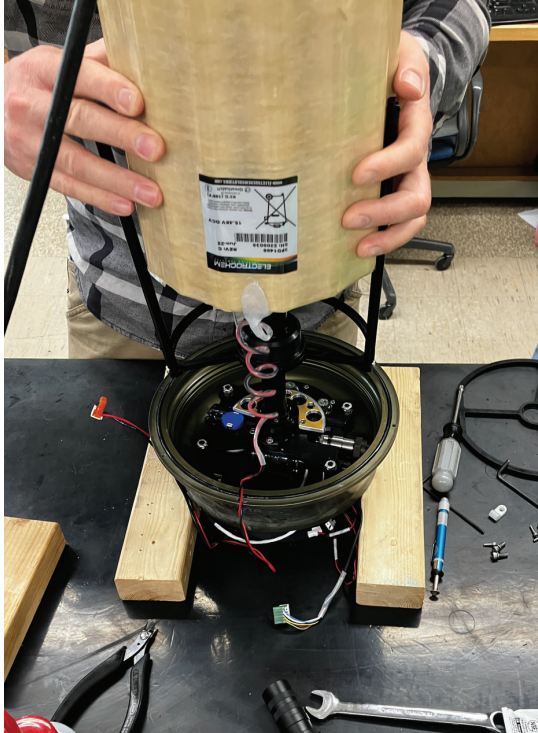
- Verify that any connectors that were removed from the tailboard and/or scicon are reattached.
- Make sure the o-ring is in place (with cleaned o-ring grooves and new, greased o-ring)
- Slide the aft endcap carefully onto the hull. Wrap cables around the mass shifter post (stinger) to keep them from snagging
- Use the mark made earlier to align the aft endcap on the hull
- Fasten all screws; use tef-gel on fasteners
- Rotate the glider so it is at a 45° angle, then reclip the cables: clip the coax first, tucking the excess in the hull using the mark made earlier as a guide for the amount of cable that should be sticking out.
- Clip the ribbon cables to the hull, being careful not to pinch them

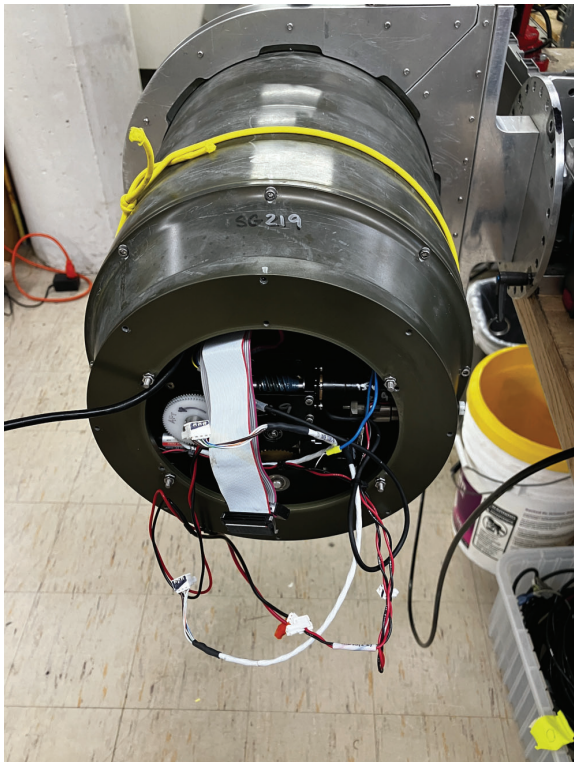


- (3) Reinstall the aft fairing, hatch covers, and sensors
- Use silicone lubricant on all connectors (both sides)
 - Take the dust cap off the DGO connector, verify there is an o-ring
 - When reinstalling the DGO, tighten snugly but do not overtighten



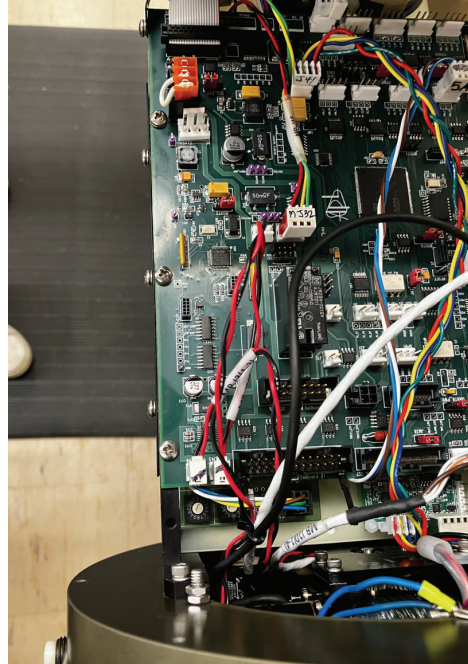
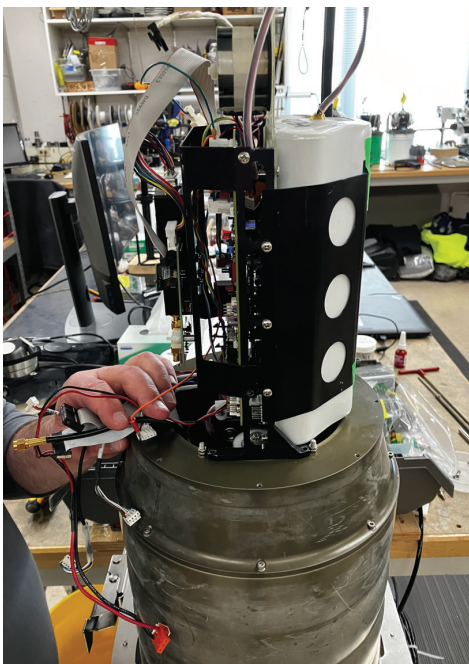
- (4) Reinstall the main battery pack
- Check the voltage: new batteries should be ~ 15.7 V
 - Run the battery through a degausser to minimize the hard-iron magnetic field (this is best practice but not critical)
 - Slide the battery onto the mass shifter: align the flat side of the battery with the brass plate and the grooves in the battery with the vertical supports. This can take some force as not all battery packs are built perfectly
 - Reattach the plate on top of the battery back
 - Feed the cable through the flanged hull section
 - Attach the battery pigtail to the gear cover: it can go on either fastener, so choose the position that fits best (e.g., wire not touching the worm gear). Don't forget the loctite.
 - Make sure the o-ring is in place (with cleaned o-ring grooves and new, greased o-ring)
 - Slide the mass shifter-battery assembly into the hull section, brass plate facing down. Make sure the cables are out of the way
 - When the two sections are a couple inches apart, feed all of the cables through, being careful not to tangle with the battery leads
 - Fasten the mass shifter hull section to the main hull section





(5) Reinstall the electronics/forward battery pack assembly

- Slide the forward battery pack into the electronics assembly. It is a tight fit.
- Apply tape to ensure that the battery doesn't slide out
- With the forward end of the glider up, fasten the electronics rails onto the hull section then tilt the glider horizontal
- Plug in (in this order): the ribbon cables, pressure sensor, pitch/roll potentiometers, and battery leads. Make sure the ribbon cables go on straight so the pins don't get bent



(6) Nose section

- Slide the nose section partway on
- Connect the nose cables to the main board: ground strap, transducer, compass
- BEFORE closing up the glider, connect to the glider with the comms cable and run through all items in the hardware menu again (as in section II):
 - o During pitch and roll moves, look into the glider with a flashlight to see how the cables respond. If cables are getting snagged or squished, adjust them as needed. This may require disassembling the glider again to adjust the cable routing
 - o While running through the hardware menu, verify that motor rates/currents haven't changed significantly; that all sensors and electronics (GPS, compass, etc) are attached and producing reasonable outputs
 - o Verify that the sensors are all plugged in and behaving correctly
- Once everything looks good, fasten the nose section



(7) Pull a vacuum:

- connect to the glider comms and run an internal pressure selftest, which will continuously output the internal pressure
- use the pressure relief valve (PRV) tool to fully tighten the PRV, then unscrew 6 turns

- Connect the vacuum to the PRV and run the vacuum until the internal pressure reads ~8.5 PSI
- Fully tighten the PRV, then unscrew 4 turns.
- Verify that the internal pressure reading holds steady. Check it again the next day.

(8) Attach the forward fairing

- Ensure that it is pushed all the way up, with no gap between the fairing sections
- Use tef-gel on all fasteners and fairing screws



V. Selftest

Run a selftest to test all hardware, sensors, electronics, and comms. Refer to user guide or piloting guide for details on reading selftest output.

VI. Ballasting

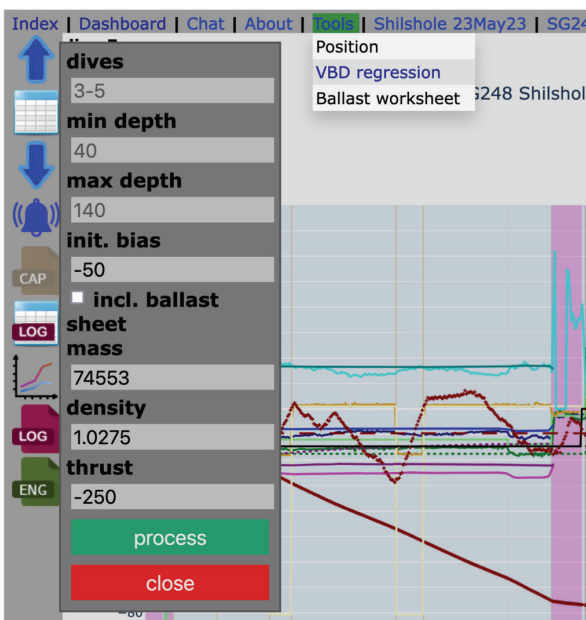
Following the refurbishment, obtain the scale mass of the glider (with all fairings, wings, rudder, antenna, and sensors attached).

Obtain an estimate of **volmax** (the glider's maximum volume, i.e., its volume with the bladder full). Typical ways of estimating volmax:

- If the configuration of sensors hasn't changed, use volmax estimated from the most recent deployment data
- If a freshwater tank large enough to accommodate the glider is available, suspend the glider on a load cell in a freshwater tank (with measured water temperature, i.e., known water density) and obtaining its weight at a given VBD position. We do this over 4 VBD positions to get better statistics.
- From a sea test. By regressing observed vertical velocity to modeled vertical velocity from several dives, a good estimate of volmax can be obtained.
 - o The basestation3 **regression tool** can be used to estimate maxvol from selected mission data:

VBD regression tool (basestation3)

Iterative regression of observed vs. modeled vertical velocity, varying flight coefficients and VBD bias. The resulting best fit gives us our best estimate of volmax.



← Dives over which to compute the regression- only use dives for which the glider flew well. We use at least 3 dives to 150 m to get best regression results.

➤ Depth range over which to compute the regression (avoid the upper ~40 m and lowest ~10 m, when the glider model isn't accurate)

← First guess of bias for the iterative regression process

← Include the ballast sheet tool in the regression output

← Scale mass - populated from the log file

➤ Target density and thrust - only used for the ballast estimate

Once an estimate of volmax is obtained, the basestation3 **ballast worksheet** can be used to estimate the target mass given a scale mass, target density and thrust, and VBD_MAX and VBD_MIN. For instance:

VBD min counts	648
VBD max counts	3962
scale mass as flown in field (g)	51657
target thrust for final ballast (g)	-150
target density for final ballast (g/cc)	1.023
antenna volume loss (cc)	150
volmax from regression of field test data (cc)	51097.06696
calculate	

Deeplighter parameters (leave blank for SG/SGX):

abs compress (model or fixed) (m ³ /dbar)	
therm expan (model or fixed) (m ³ /°C)	
ref temperature (°C)	
apogee temperature (°C)	
apogee pressure (dbar)	
calculate Deeplighter	

comments

Ballast worksheet

Automatically populated from the most recent log file
← actual mass of glider

Typical values we use for testing in Puget Sound
For open ocean, typical target values are -250 g thrust, 1.0275 g/cc density

Volume that is out of the water when the glider is at the surface. Antenna is always 150cc.

volmax - from tank or from regression to field data

This button calculates the final ballast results from the above parameters

Ignore for SG/SGX

Optional box for adding comments
(for record keeping purposes)

Final ballast results

lead to add for final ballast (g)	-72.9263856	Calculated final ballast and lead to add/remove
target mass for final ballast (g)	51584.07361	
final scale mass after adjustments (g)		Optional - enter the actual final mass (for record keeping)
predicted volmax at final ballast (cc)	51090.61101	Good starting place for trimming the glider (typically ~200 counts higher than the actual C_VBD if ballasted on ~150m flights for open ocean operating at 1000m) The glider will struggle to surface if the surface is less dense than this
predicted C_VBD at final ballast	3364.25207	
minimum surface density (g/cc)	1.012631623	
maximum buoyancy (cc)	666.296634	
neutral stroke (%)	81.96294734	

The “final ballast results” section gives the amount of lead to add in order to achieve the desired ballast.

Lead placement:

When ballasting, place (or move) lead to achieve the desired trim following these general principles:

(1) Pitch.

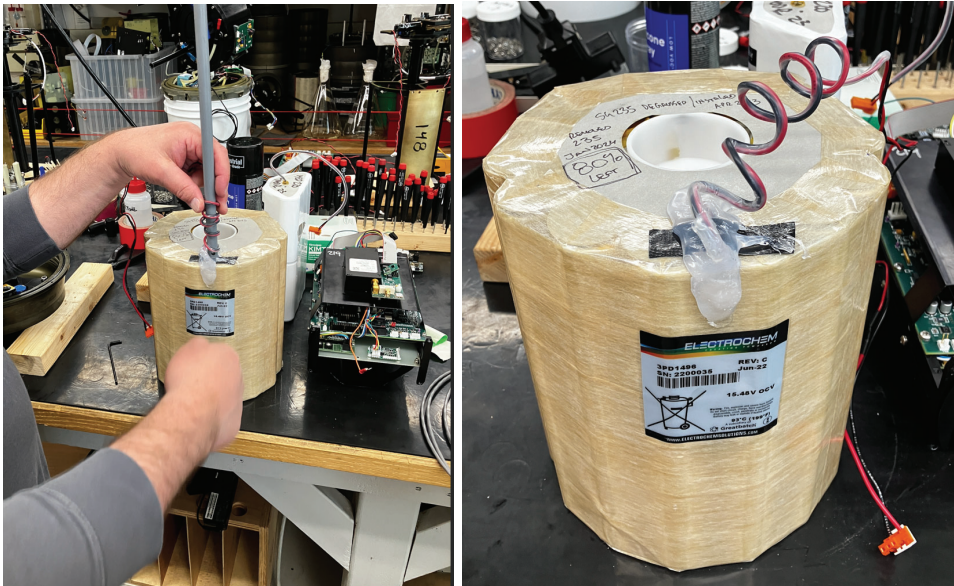
- We aim for *approximately* -60° pitch when the glider is at the surface (though in practice the surface angle is often shallower than this) and ±18° pitch when the glider is diving/climbing. If the glider’s pitch is too flat, lead can be placed toward the nose. If the glider’s pitch is too steep, lead can be placed aft.
- We aim very roughly for a C_PITCH of 2400 AD counts (± a few hundred AD counts): this pitch center gives the glider enough throw both forward and aft. If C_PITCH is much lower than this (i.e., the pitch center is too far forward, so the glider can’t pitch forward enough and the surface angle is too flat), place lead toward the nose. Similarly, if C_PITCH is too high, place lead aft.

(2) Roll.

- Vehicle rolls should generally be symmetric to port/starboard, and very roughly around $\pm 18^\circ$. If the vehicle is not rolling enough to one or both sides, lead can be placed on the sides (rather than on the glider's belly): this destabilizes the glider, making it turn more easily
- Note that the SGX mass shifter can roll up to $\pm 80^\circ$ (compared to $\pm 40^\circ$ for the original SG): this additional throw is needed to achieve sufficient vehicle roll in order to turn. To achieve symmetric roll on SGX, lead placement is typically asymmetrical

VII. Battery notes

- (1) It is important that the battery leads have a nice pigtail so that they stretch and retract when the mass shifter moves the battery. Coil the heat-shrunk section of battery leads around a stick apply heat with a heat gun to form or refresh the pigtail:



- (2) If it is necessary to replace the battery pack's fuse, it can be located under the cardboard wrapping, on the long side of the battery pack around where the leads are. The fuses for 15V SG and SGX battery packs are as follows:

Battery pack (part number)	Fuse
Seaglider/SGX 15V main pack (3PD1496)	Littlefuse 0251020 (20 A)
Seaglider 15V forward pack (3PD1503)	Littlefuse 0251010 (10 A)
SGX 15V forward and aft packs (3PD1632, 3PD1633)	Littlefuse 0251015 (15 A)